INCH-POUND

MIL-I-5184B

10 July 1989
SUPERSEDING
MIL-I-5184A
3 OCTOBER 1951

MILITATY SPECIFICATION

INDICATOR, POSITION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of 28V dc, wing flaps position indicator.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Method Of
MIL-T-5583	Transmitter; Position, 28 Volt DC
MIL-I-7062	Indicator, Postion, Control Surfaces, 28 Volt DC, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-2073-1	DoD Material Procedures For Development And Application Of Packaging Requirements.
MS28023	Indicator-Position, Wing Flaps, Percent Marking
MS33638	Case, Instrument, Flange Mount, Aircraft

Air Force-Navy Aeronautical

AN5786	Transmitter-Position, 28 Volt DC
AN5787	Transmitter-Position, Gearless

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM) ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in Specification MIL-I-7062 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

a. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design and construction. The indicator shall be designed for operation with a position transmitter in accordance with Specification MIL-I-5583. The indicator pointer shall travel approximately 180 mechanical degrees for 90 degrees rotation of a transmitter conforming to Drawing AN5786 or 180 degrees rotation of a transmitter

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conforming to Drawing AN5787.

3.2.1 Resistance. The direct-current resistances of the indicator measured between the connector pins A, B, and C shall be 475 ±100 ohms. Deviation from the average of the resistance values between the three coils shall not exceed ±1.50 percent.

3.2.2 Pointer overlap. The pointer length shall be such that the pointer tip will extend into the scale a distance equal to 0.333 to 0.67 of the length of the shortest graduations.

3.2.3 Electrical Connector. A connector as shown on Standard MS28023 shall be furnished, unless otherwise approved for hermetically sealed instruments by the Contracting Service.

3.2.4 Cover Glass Position. The distance between the outer surface of the cover glass and the front edge of the case shall not exceed 0.031 inch.

3.2.5 Case. The case shall be in accordance with Standard MS28023. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator.

a. The design of the indicator case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the Contracting Service. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

b. The case shall be filled with helium of at least 98-percent purity, free of dust particles, and containing not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The pressure of the helium in the case shall be 1 atmosphere ±psi as referenced to an absolute vacuum.

3.2.6 Dial. The dial shall conform to figure 1. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. The diameter of the dial shall be not less than 1.75 inches when measured across the outside ends of the graduations.

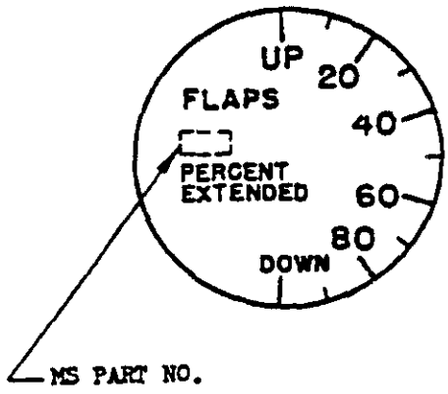
3.3 Markings. The dial markings shall be as shown on figure 1, and shall be painted with a pale yellow, fluorescent luminescent material in accordance with MIL-L-3891. The painted surfaces shall experience no degradation as a result of the environmental tests in section 4 of this specification.

3.4 Outline Dimensions. The outline dimensions of the indicator shall conform to Standard MS28023.

3.5 Identification of product. The nameplate shall be marked with the following information:

INDICATOR, POSITION
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

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MARKINGS	HEIGHT OR LENGTH INCHES ±0.015	WIDTH OF LINE OR GRADUATION INCH ±0.005	MATERIAL OR FINISH
Numerals 20, 40, 60, 80, and lettering "UP"	0.156	0.010	
Lettering "FLAPS"	0.125	0.010	
Lettering "DOWN"	0.109	0.010	Flourescent
Lettering "PERCENT EXTENDED"	0.093	0.010	Luminescent
Major Graduations	0.187	0.031	
Minor Graduations	0.043	0.020	
Shaded Porton of Pointer	———	———	
MS Part No.	0.062	0.020	Durable
Background of Dial	———	———	Dull Black

FIGURE 1. Dial Markings.

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3.6 Recycled and reclaimed material. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 General. Methods of sampling, inspection, and tests shall be as specified in Specification MIL-I-7062 and as follows:

4.2.1 Scale Error Test. The scale error shall be read from the test transmitter scale with the indicator pointer on each major graduation of the indicator dial.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation Application. Preservative shall not be used.

5.1.1.4 Unit Packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for

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the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of the indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flaps position indicator covered by this specification is intended to indicate the position of the wing flaps in percent of travel from the up position. The position transmitters used with this indicator are connected to the flap actuating mechanism so that the transmitter shaft is rotated 90 or 180 mechanical degrees as applicable for 100-percent flap travel. The indicator pointer rotates approximately 180 degrees to indicate the full down position.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c Levels of preservation and packing (see 5.1 and 5.2).

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6.3 Subject term (key word) listing.

Indicator
Transmitter

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

AIR FORCE -99
Army -AV
Navy -AS

Preparing Activity

AIR FORCE -71

Project No.
6610-0288

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-5184B	Indicator, Position

NAME OF SUBMITTING ORGANIZATION

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

b. ADDRESS *(Street, City, State, ZIP Code)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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